

AN APPROACH TO FLOODING PREDICTION USING GIS AND TIME SERIES DECOMPOSITION: An example of flooding in Mohawk River, New York

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Flood caused by Storm



***Route 145 in Middleburgh, NY (28/8/11;
photo by Times Union)***



***Mr. Jim Duggan makes his way back up North street
in Schenectady, NY (28/8/11; photo by Times Union)***



***North Ferry Street in Schenectady, NY (28/8/11;
photo by Paul Buckowski/Times Union)***



Mohawk River – Flood 29/08/11

*Flood caused by **Ice jam***



A photograph of the Mohawk River in winter. The river is partially frozen with ice and snow. In the background, a bridge spans the river, and a line of bare trees is visible on the opposite bank. In the foreground, there are large trees on the right and a rocky, snow-dusted bank on the left.

Mohawk River
New York

*What causes an **Ice jam**?*

- After a cold weather and warmer temperatures



Mohawk River, NY – January 2

1. Snow melt increases river's flow
2. Ice sheets break apart, and move downstream
3. Ice chunks pile up

*What causes an **Ice jam**?*

- After a cold weather and warmer temperatures



Mohawk River, N

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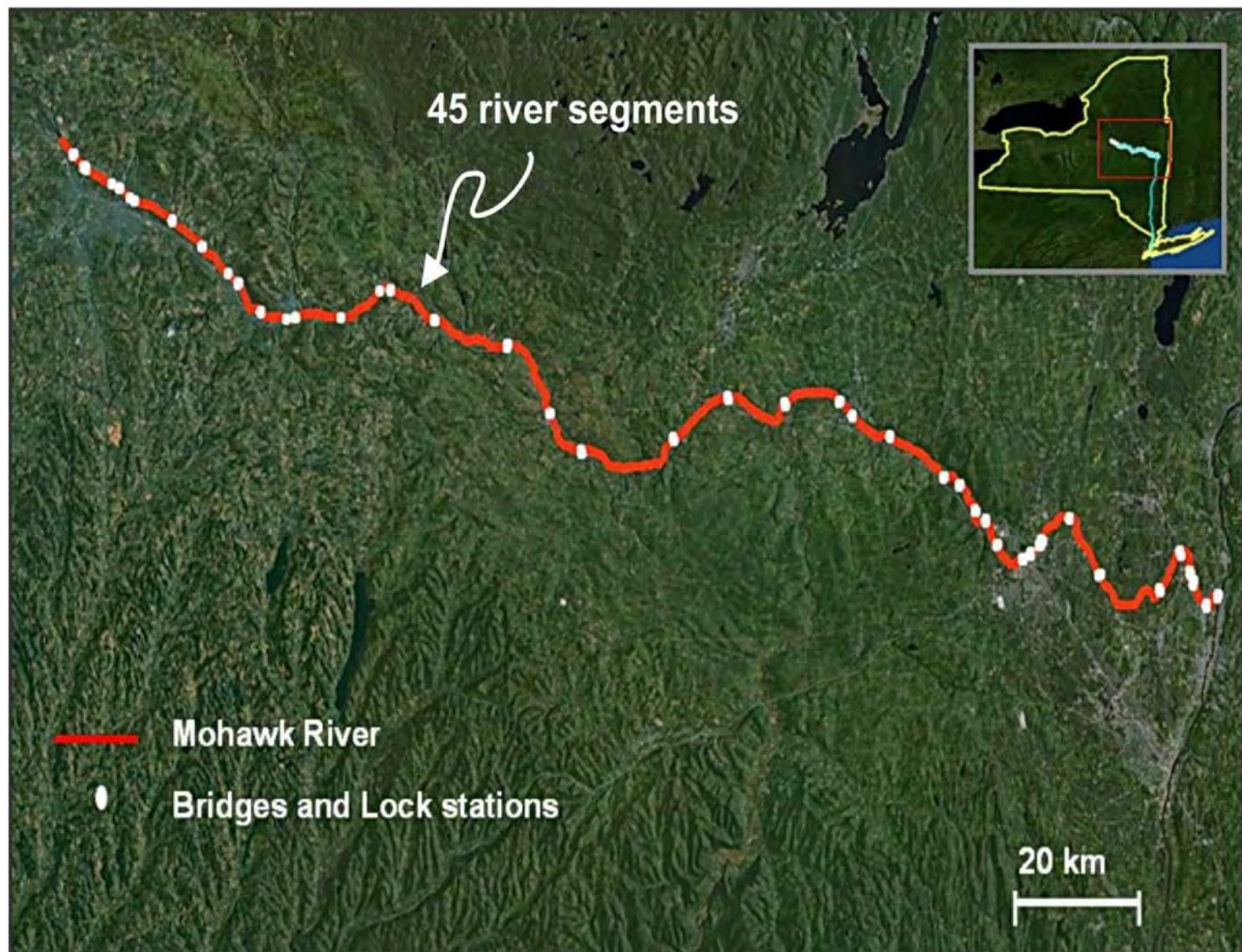
Mohawk River, N

1. Snow melt increases river's flow
2. Ice sheets break apart, and move downstream
3. Ice chunks pile up & and ice jam forms in restricted water flow areas (e.g. bridges)

January 2010







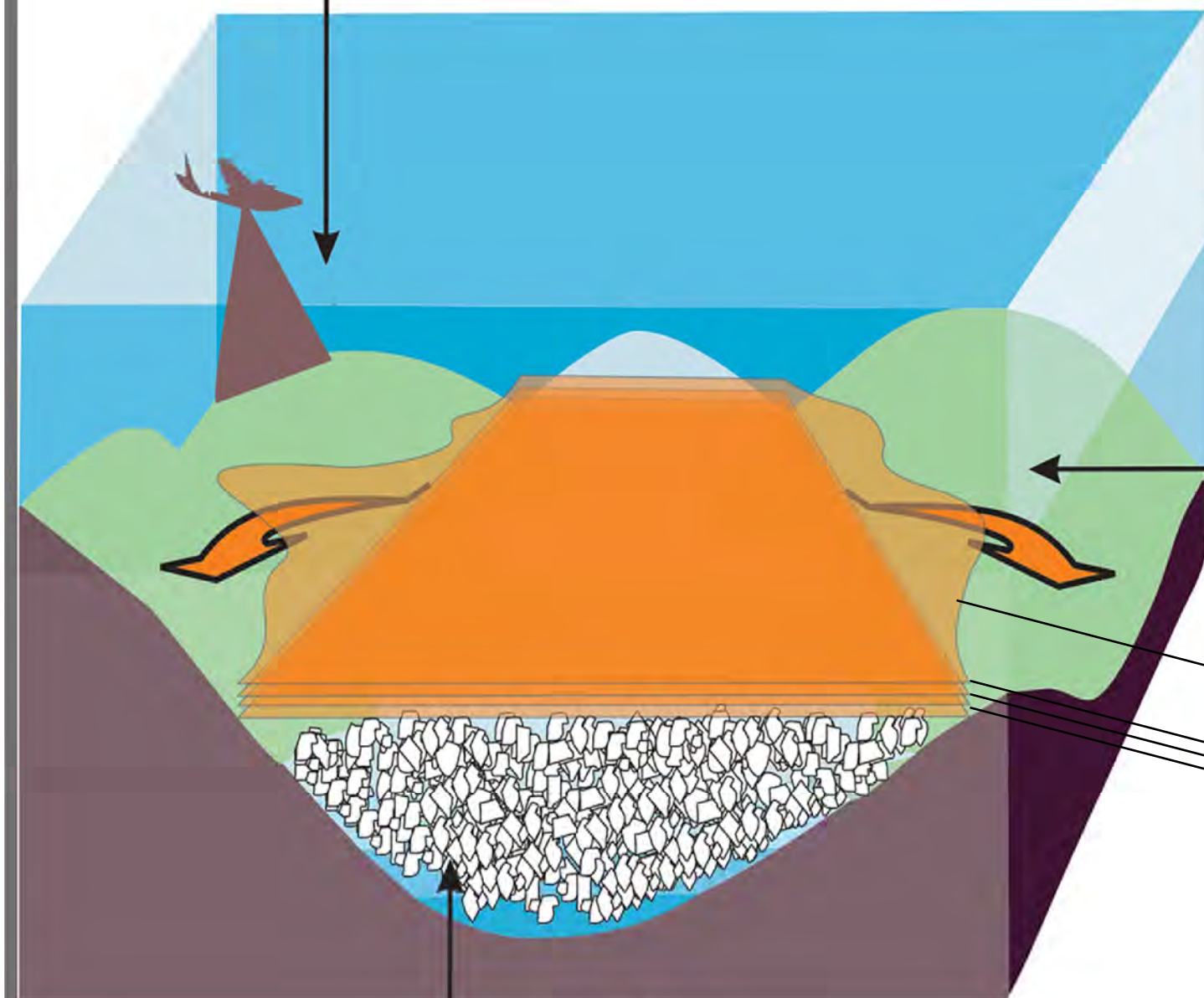
LiDAR DEM

Backed up
water fills
the floodplain

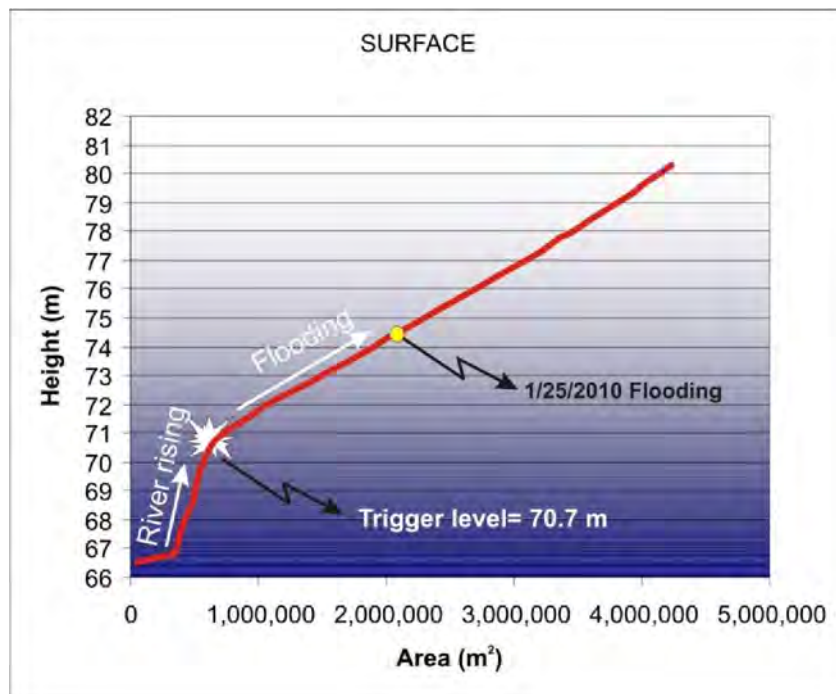
FTL

.
. 50.7 m
. 50.6 m
. 50.5 m
. .
. 0.1 m

Ice dam

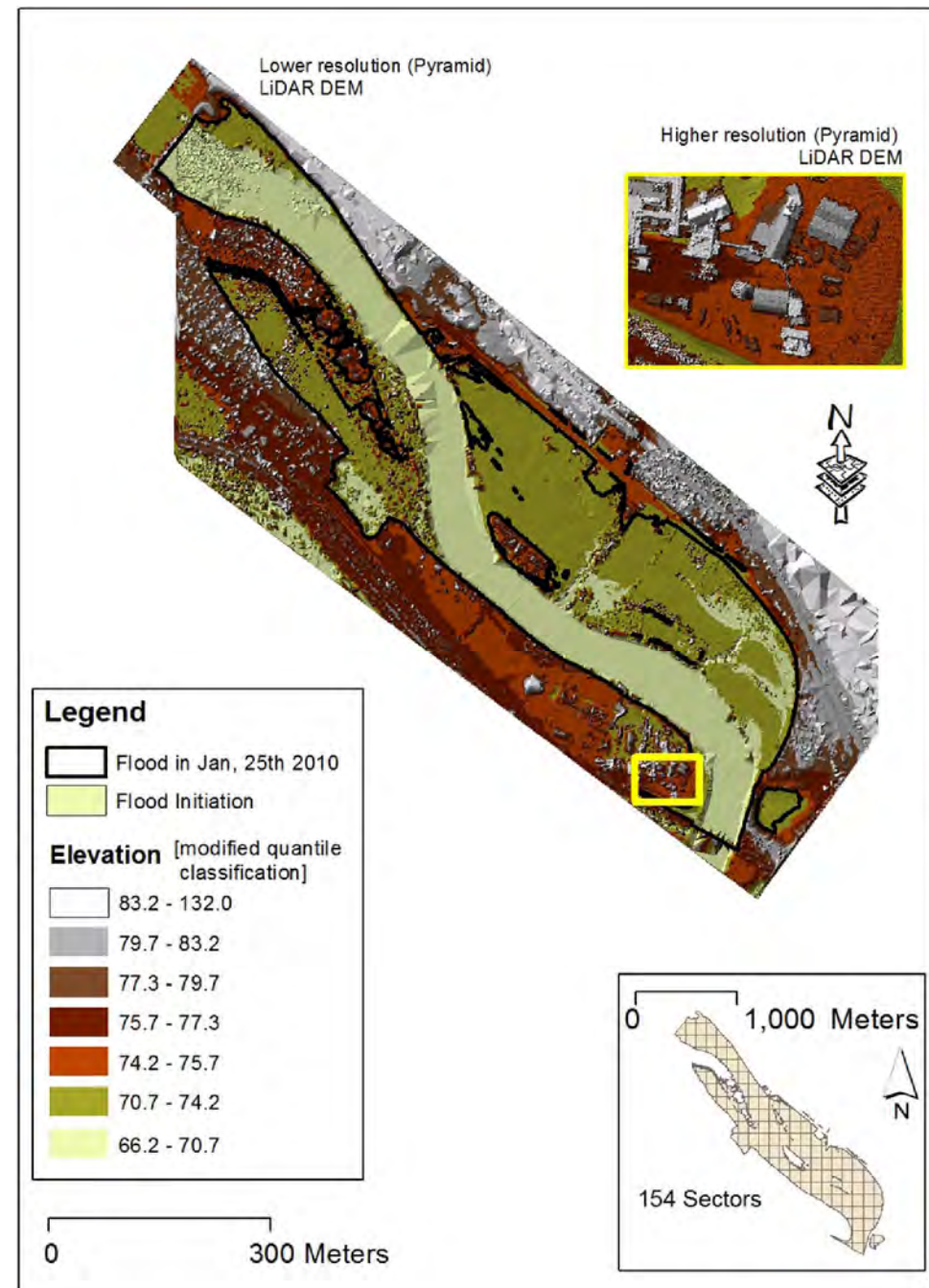


Simulation & Prediction



(*Foster, J., Marsellos, A., Garver, J., 2011)

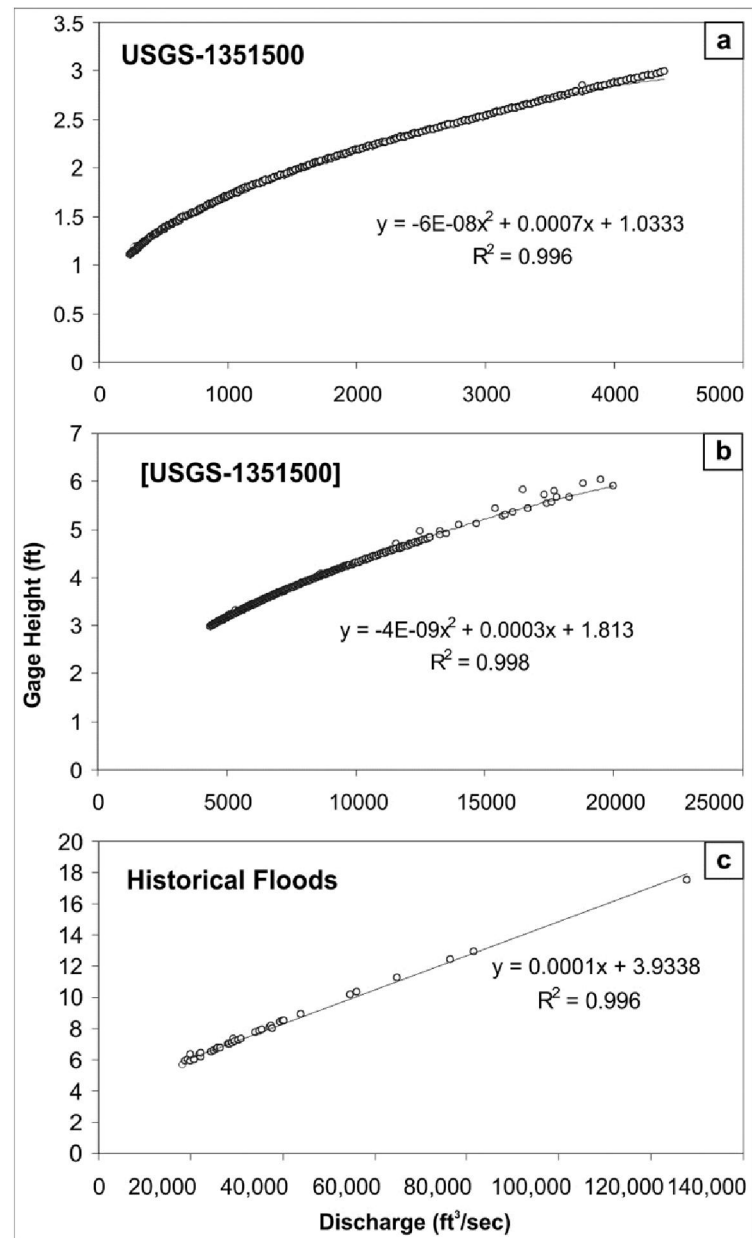
*undergraduate supervised student



Time Series Data

- ☐ Water Discharge
- ☐ Ground water level
- ☐ Precipitation
- ☐ Temperature
- ☐ Atmospheric Pressure
- ☐ Tide

Water discharge and height gage association



Decomposition of time series

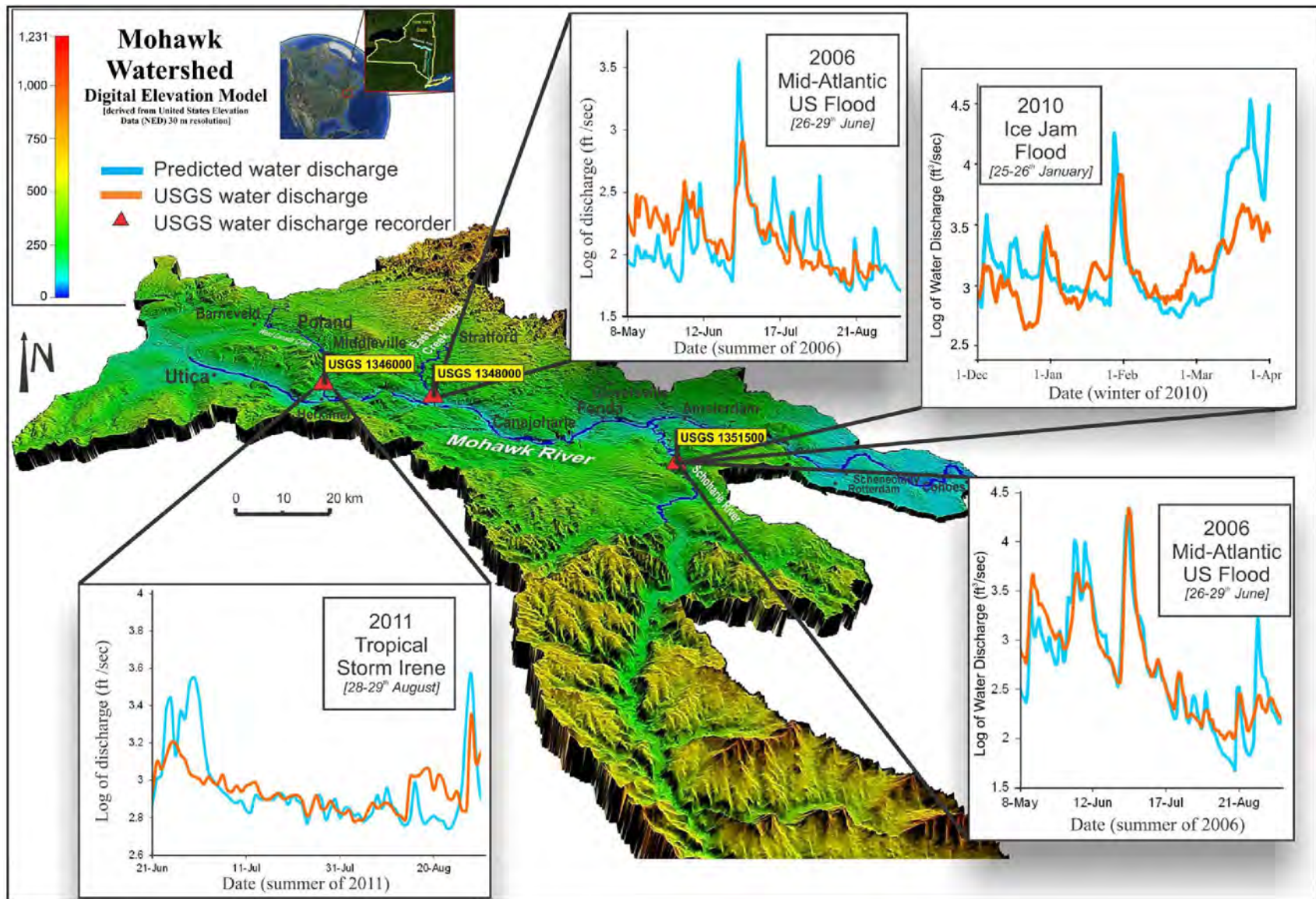
$$X(t) = L(t) + Se(t) + S(t)$$

- $X(t)$: original time series
- $L(t)$: long term trend component
- $Se(t)$: seasonal component
- $S(t)$: short term component

Decomposition of Time Series

$$X(t) = L(t) + Se(t) + S(t)$$

- The long term component is related to the long term changes
- The seasonal component describes the year-to-year fluctuations
- The short term component is related with the short term variations



CONCLUSIONS

- LiDAR data and GIS techniques allow the simulation of the water level which will trigger flood in the rivers.
- Decomposition of the time series model predicts the initiation time of flooding.
- This methodology can be applied for real-time spatial analysis for disaster management or preventing a flooding hazard in a river such as in the Mohawk River in New York.

Thank you